

Density based clustering

- based on notion of density
- basic idea is to continue growing the given cluster as long as the density in the neighbourhood exceeds some threshold i.e., for each datapoint within a given cluster, the radius of a given cluster has to contain at least a minimum no. of points

Major features

- discover clusters of arbitrary shape
 - handle noise
 - need density parameters as termination condⁿ
- eg- dbscan, optics, denclue

DBSCAN

- density based spatial clustering of applications with noise
- robust to outliers
- does not need the no. of clusters beforehand
- needs 2 parameters:

ϵ - radius of the circle to be created around each data point to check the density

minPoints - minimum no. of data points in an ϵ neighborhood of that point

Density at a point P

No. of points within a hypersphere of radius ϵ around P.

Dense Region

A hypersphere of radius ϵ that contains atleast 'minPoints' no. of points.

• DBSCAN creates a circle of ϵ radius around every data point and classifies them into core point, boundary point and noise point.

i) A data point is a core point if the circle around it (with ϵ radius) contains at least 'min Points' in its radius.

ii) Every core point belongs to dense region.

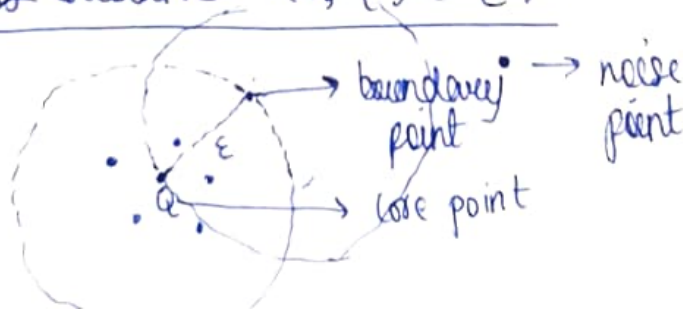
ii) A point P is said to be boundary point if it is not a core point.

a) P has $<$ 'minPoints' points in ϵ radius

b) But P belongs to a neighborhood of Q where Q is a core point.

means distance $(P, Q) \leq \epsilon$.

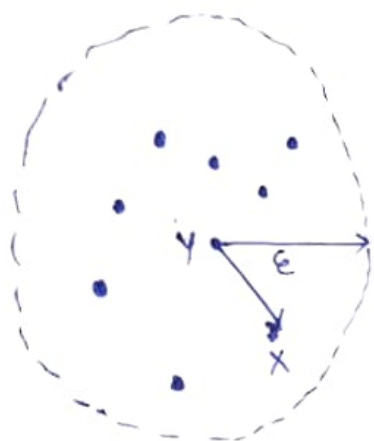
min points
= 5



Directly Density-Reachable

A point x is directly density-reachable from point y w.r.t. ϵ , minPoints, if

- x belongs to the neighborhood of y , i.e., $\text{dist}(x, y) \leq \epsilon$
- y is a core point



Thus, x is directly density-reachable from y , but vice-versa is not true because x is not a core point.

Density Connected - Points belonging to same cluster

Algo of DBSCAN

Step 1: Label core point and noise point

- Select a random starting point, say 0
- Identify neighborhood of this point 0 using the radius ϵ
- Count the no. of points, k , in this neighborhood including point 0
- if $k \geq \text{minpts}$ then mark 0 as core point
- else it will be marked as noise point
- Select a new unvisited point and repeat above steps

Step 2: check if noise point can become boundary point

- i) If noise point is directly density reachable (that is within the boundary of radius ϵ from the core point), mark it as boundary point and it will form the part of cluster
- ii) otherwise it will remain noise point

Parameter selection

The value of minpoints should be atleast one quarter than the no. of dimensions of the dataset i.e.,

$$\boxed{\text{minpoints} \geq \text{dimensions} + 1}$$

	x	y
P1	3	7
P2	4	6
P3	5	5
P4	6	4
P5	7	3
P6	6	2
P7	7	2
P8	8	4
P9	3	3
P10	2	6
P11	3	5
P12	2	4

$$\epsilon = 1.9 \quad \text{minPts} = 4$$

euclidean distance matrix (symmetric)

P ₁	0												
P ₂	1.41	0											
P ₃	2.83	1.41	0										
P ₄	4.24	2.83	1.41	0									
P ₅	5.66	4.24	2.83	1.41	0								
P ₆	5.83	4.47	3.16	2.00	1.41	0							
P ₇	6.40	5.00	3.61	2.24	1.00	1.00	0						
P ₈	5.83	4.47	3.16	2.00	1.41	2.83	2.24	0					
P ₉	4.00	3.16	2.83	3.16	4.00	3.16	4.12	5.10	0				
P ₁₀	1.41	2.00	3.16	4.47	5.83	5.66	6.40	6.32	3.16	0			
P ₁₁	2.00	1.41	2	3.16	4.47	4.24	5	5.1	2	1.41	0		
P ₁₂	3.16	2.83	3.16	4	5.10	4.47	5.39	6	1.41	2	1.41	0	
	P ₁	P ₂	P ₃	P ₄	P ₅	P ₆	P ₇	P ₈	P ₉	P ₁₀	P ₁₁	P ₁₂	

$P1: P2, P10$ $P2: P11, P3, P1$ $P3: P4, P2$ $P4: P5, P3$
 $P5: P8, P1, P6, P4$ $P6: P5, P1$ $P7: P6, P5$ $P8: P5$
 $P9: P12$ $P10: P1, P11$ $P11: P2, P10, P12$ $P12: P11, P9$

Point	Status
P1	noise → border
P2	core
P3	noise → border
P4	noise → border
P5	core
P6	noise → "
P7	" → "
P8	" → "
P9	" → <u>only one</u>
P10	" → " <u>noise point</u>
P11	core
P12	noise → "